DEVELOPMENT ROADMAP — VRU TRADING BOT
Horizon Tech — VRU Trading Bot Research
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CURRENT STATE (as of May 2026)

- VRU v23 in active development: explicit 6-slot register architecture
- Training on arithmetic/synthetic curriculum on Vast.ai RTX 5090 (instance 32400733)
- Architecture proven on: chaotic sequences (Lorenz), multi-stream telemetry, long sequences
- Output head: character-level softmax for arithmetic (must be replaced for trading)
- No trading-specific training data pipeline yet

PHASE 1 — ARCHITECTURE PIVOT (Weeks 1–4)

Goal: Retarget v23 from arithmetic reasoning to order flow classification

Tasks:
1. Replace output head
   - Remove: character-level softmax
   - Add: 3-class linear projection + softmax + confidence score
   - Loss: weighted cross-entropy

2. Replace input layer
   - Remove: character/token embedding
   - Add: continuous feature vector (price + options + order flow streams)
   - Add: normalization layer per stream

3. Synthetic order flow data generator
   - Simulate institutional VWAP execution patterns
   - Inject into base price series with noise
   - Auto-label: accumulation / distribution / flat
   - Match v23 curriculum harness structure (mastery gate, JSONL audit)

4. Validate on synthetic data
   - Target: >85% classification accuracy on holdout before moving to real data
   - Stress test at 5,000+ timestep sequences

PHASE 2 — DATA PIPELINE BUILD (Weeks 3–8, overlaps Phase 1)

Goal: Live third-party data ingestion

Tasks:
1. API integrations

   - Unusual Whales: options flow websocket
   - Quiver Quant: dark pool prints, congressional trades
   - Alpaca or Polygon.io: 1-minute OHLCV + Level 2

  2. Normalization layer
     - Rolling Z-scores per stream
     - Lag features: t-1, t-5, t-10
     - Cross-stream correlation features

  3. Historical data pull
     - Minimum 2 years of all streams
     - Align timestamps across sources (different latencies)
     - Store in structured format for training

  4. Label generation
     - Use 13F delta as ground truth for confirmed institutional moves
     - Build labeling pipeline: confirmed move → backfill labels on options/dark pool

PHASE 3 — TRAINING ON REAL DATA (Weeks 6–14)

Goal: Train VRU on historical order flow data

Tasks:
  1. Walk-forward validation setup
     - NO random train/test split (look-ahead bias)
     - Train on months 1–18, validate on months 19–24, repeat rolling

  2. Hyperparameter optimization
     - Hidden size: start at 128, scale to 256 if compute allows
     - Sequence length: 1,440 (1 day) minimum, test 4,320 (3 days)
     - Confidence threshold tuning: optimize on validation Sharpe, not accuracy

  3. Regime robustness testing
     - Test on: bull market, bear market, high vol, low vol, trending, ranging
     - VRU must generalize across regimes — not just fit one market condition

PHASE 4 — PAPER TRADING (Weeks 12–24)

Goal: Live signal generation, simulated execution

Setup:
  - Real-time data pipeline running
  - Signal generated every 1-minute bar close

- Simulated execution at next bar open (realistic fill assumption)
- Track: signal accuracy, Sharpe, drawdown, win rate by confidence bucket

Minimum gate before Phase 5:
- 90 calendar days of paper trading
- Sharpe > 1.0 on paper
- Max drawdown < 15% on paper
- Win rate > 55% on high-confidence signals (>0.75)

PHASE 5 — LIVE TRADING

Starting capital: $5,000–$25,000 (minimum viable)
Scale trigger: 6 months live track record matching paper performance

Ongoing:
- Weekly model retraining on rolling data window
- Regression detection monitoring (circuit breaker halts if model degrades)
- JSONL audit trail on all signals, positions, P&L
- Quarterly review: add new data sources, adjust confidence thresholds

COMPUTE REQUIREMENTS

Training: Vast.ai RTX 5090 (instance 32400733) — current setup adequate
Inference: Much lighter — can run on smaller GPU or even CPU for 1-minute signals
Data storage: Google Drive /dppu_vru/ — extend with dedicated database for tick data

KEY RISKS
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1. Label noise: 13F-confirmed labeling is quarterly; real-time labels are inferred
2. Regime change: model trained in one regime may not generalize
3. Data vendor reliability: API outages affect live trading
4. Overfitting: walk-forward validation mitigates but doesn't eliminate
5. Execution slippage: limit orders help but wide spreads can erode signal value

MITIGATION:
- Conservative confidence thresholds (no trade below 0.60)
- Hard stop-losses with no model override
- Quarter-Kelly position sizing until 6-month track record established
- Paper trading minimum before any real capital